CYTECH ACADEMY

Linux & Security Tools Master Guide

Complete Study Notes - beginner friendly, exam and lab ready.

(c) CyTech. For enrolled students. Unauthorized attack is illegal - practice only in labs.

Table of Contents

1. Linux for Security Work

2. Recon & Scanning Toolkit

3. Exploitation & Web Toolkit

4. Passwords, Wireless & Forensics

5. Reverse Engineering Tools

Linux for Security Work

Kali Linux is the standard pentest distro; Parrot OS is a lighter alternative. Learn the shell deeply: find, grep, awk, sed, chmod/chown, ps, netstat/ss, curl, wget, tar, systemctl, journalctl. Permissions (rwxrwxrwx, SUID, sudo) are a core privilege-escalation topic.

Build a lab: VirtualBox/VMware + Kali + Metasploitable2 + DVWA on a host-only network. Snapshot before every experiment.

Recon & Scanning Toolkit

Nmap: -sS stealth SYN, -sV version, -sC default scripts, -A aggressive, --script vuln, -oN/-oX output. Masscan for huge ranges (lab only). theHarvester & sublist3r for OSINT/subdomains.

Web: ffuf/gobuster for directories (ffuf -w wordlist -u URL/FUZZ), whatweb/wappalyzer for fingerprinting, nikto for known misconfigs.

Exploitation & Web Toolkit

Metasploit: modules for thousands of CVEs; meterpreter gives a powerful post-exploitation shell. sqlmap automates SQLi end-to-end including DB dump. Hydra/medusa brute-force logins (ssh, ftp, http forms).

Burp Suite / OWASP ZAP: intercept, repeater, intruder, scanner. Configure your browser proxy to 127.0.0.1:8080 and install the CA cert to inspect HTTPS.

Passwords, Wireless & Forensics

John the Ripper: john --wordlist=rockyou.txt hashes.txt. Hashcat uses your GPU: hashcat -m 0 hashes.txt rockyou.txt (mode 0 = MD5; 1000 = NTLM; 1800 = sha512crypt). Identify hash types with hashid.

Wireless: aircrack-ng suite (airodump capture, aireplay deauth, aircrack crack WPA handshake) - only on your own network. Forensics: Autopsy (disk), volatility (memory), binwalk (firmware/embedded files), exiftool (metadata).

Reverse Engineering Tools

Ghidra (decompiler), IDA Free, cutter/radare2, x64dbg, gdb+gef, ltrace/strace (library/system call tracing), upx (packer), Detect It Easy (packer identification). Practice on crackmes.one - legal and addictive.